

Mitchell B Slapik

MD/PhD Candidate & NIH Fellow, McGovern Medical School, mslapik@gmail.com

Education

2027	McGovern Medical School, <i>Houston, TX</i> Medical Scientist Training Program	MD
2025	Graduate School of Biomedical Sciences <i>Houston, TX</i> , Neuroscience	PhD
2017	Johns Hopkins University, <i>Baltimore, MD</i> Post-Baccalaureate Premedical Program	
2014	Swarthmore College, <i>Swarthmore, PA</i> With High Honors in Philosophy and Linguistics	BA
2013	University of Oxford, <i>Oxford, UK</i> Study Abroad: Philosophy of Mind	

Publications

2026	R. Milton*, M. Slapik* , S. Egranov, et al. "Locomotor Activity Enhances Environmental Representation by Boosting Visuo-frontal Communication." Under review at <i>Nature Neuroscience</i> . M. Slapik , H. Shouval. "Simulated Complex Cells Contribute to Object Recognition through Representational Untangling." <i>Neural Computation</i> .
2025	M. Slapik . "Computational Evidence for an Inverse Relationship between Brain and Retinal Complexity." <i>Journal of Vision</i> .
2022	M. Joyce, P. Nadkarni, S. Kronemer, [...], M. Slapik , et al. "Quality of Life Changes Following the Onset of Cerebellar Ataxia: Symptoms and Concerns Self-reported by Ataxia Patients and Informants." <i>Cerebellum</i> .
2020	O. Morgan, M. Slapik , K. Iannuzzelli, et al. "The Cerebellum and Sequencing in Motor and Cognitive Domains: Evidence from Cerebellar Ataxia." <i>Cerebellum</i> . S. Kronemer, M. Slapik , J. Pietrowski, et al. "Neuropsychiatric Symptoms as a Reliable Phenomenology of Cerebellar Ataxia." <i>Cerebellum</i> .
2018	M. Slapik , S. I. Kronemer, O. Morgan, et al. "Visuospatial Organization and Recall in Cerebellar Ataxia." <i>Cerebellum</i> .

* denotes equal contribution.

Research experience

- 2021 – Now **Graduate Research Assistant, Dragoi Lab**
McGovern Medical School, *Houston, TX*
Investigate parallels between biological and artificial intelligence in visual processing
- Analyzed neural activity in V4 and dlPFC during free exploration, and quantified inter-areal communication using reduced-rank regressions and spike-timing coordination
 - Studied how early visual computations contribute to representational untangling in biological and artificial networks.
 - Investigated the relationship between retinal and brain complexity using convolutional neural networks
 - Designed optimal stimuli for neurons in visual cortex using image generators and optimizers.
- 2016 - 2019 **Research Assistant, Marvel Lab**
Johns Hopkins School of Medicine, *Baltimore, MD*
Examined the cognitive and emotional effects of cerebellar ataxia
- Designed new cognitive tasks assessing visuospatial skills, gestalt processing, sequence learning and verbal encoding
 - Administered cognitive tasks, emotional questionnaires and motor tests to ataxia patients and controls
 - Analyzed behavioral and questionnaire data, contributed to peer-reviewed manuscripts and presented findings to researchers, clinicians and patient communities.

Fellowships and awards

- 2023 **National Research Service Award (F30)**
National Institutes of Health: National Eye Institute
- 2022 **Clinical and Translational Science Predoctoral Fellowship (TL1)**
National Institutes of Health: National Center for Advancing Translational Sciences
- 2024 **Dean's Research Scholarship**
McGovern Medical School
- 2023 **George M. Stancel Fellowship in the Biomedical Sciences**
McGovern Medical School
- 2023 **Best Work-In-Progress Talk**
Department of Neurobiology and Anatomy, McGovern Medical School

2022, 2023, 2024	Osborne Endowed Scholarship in the Neurosciences Department of Neurobiology and Anatomy, McGovern Medical School
2018	Travel Award for 8 th Annual Ataxia Investigator's Meeting The National Ataxia Foundation
2014	High Honors in Philosophy and Linguistics Swarthmore College Honors Program

Presentation

Selected talks

2024	M. Slapik. "Retinal Complexity Varies Inversely with Brain Complexity in a Computational Model." Post-candidacy Talk at Fall Neuroscience Retreat for UTHealth. Houston, TX.
2023	M. Slapik, H. Shouval. "Unshattering Dimensionality." Post-candidacy Talk at Fall Neuroscience Retreat for UTHealth. Cleveland, TX.
2022	M. Slapik, A. Andrei, S. Khan, et al. "Optimal Stimuli as a New Method to Investigate Neural Networks." Pre-candidacy Talk at Fall Neuroscience Retreat for UTHealth. Cleveland, TX. M. Slapik, O. Morgan, J. Creighton, et al. "Timing and Sequencing in Cerebellar Ataxia." Nanosymposium talk accepted for presentation at: Society for Neuroscience San Diego, CA. M. Slapik, O. Morgan, C. Marvel. "Language Abilities in Cerebellar Ataxia." Presentation to the faculty and staff of the Johns Hopkins Ataxia Clinic, Baltimore, MD.
2017	M. Slapik, S. Kronemer, O. Morgan, et al. "Visuospatial Organization and Recall in Cerebellar Ataxia." Talk presented at: Sensorimotor Day, Johns Hopkins University, Baltimore, MD.

Selected posters

2024	M. Slapik, X. Niu, A. McConnell, et al. "Executive Regions Encode Movement of Individual Body Parts in Macaque Monkeys." Society for Neuroscience. Chicago, IL. M. Slapik, H. Shouval. "The Visual System Relies on Organized but Low-Dimensional Representations for Object Recognition." Spring Neuroscience Retreat for UTHealth, Houston, TX.
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2023	M. Slapik , A. Andrei, S. Khan, et al. "A Deep Learning Approach to Naturalistic Surround Modulation." Society for Neuroscience. Washington, D.C.
2022	M. Slapik , A. Andrei, S. Khan, et al. "Deep Networks Design Optimal Stimuli for Early Visual Cortex." Society for Neuroscience. San Diego, CA.
2020	M. Slapik , S. Patwardhan, R. Costa, et al. "Using Machine Learning To Classify Feeding Behavior in Aplysia." American Physician Scientists Association. Houston, TX.
2018	M. Slapik , J. Pietrowski, O. P. Morgan, et al. "A Characterization of Language Impairment in Cerebellar Ataxia." The National Ataxia Foundation's 8 th Ataxia Investigator's Meeting. Philadelphia, PA.
2017	M. Slapik , S. Kronemer, J. Mandel, et al. "Visuospatial Processing and Strategy Formation in Cerebellar Ataxia." Society for Neuroscience, Washington, D.C.

Clinical and community service

2021 – Now	Volunteer Counselor Crisis Text Line, <i>Houston, TX</i> • Provide crisis support via text to individuals experiencing suicidal ideation, self-harm urges, anxiety, depression, and acute psychosocial stressors • Connect texters to personalized resources related to depression, anxiety, substance use, housing insecurity, and gender/sexual identity
2021 – Now	Shadowing, Department of Psychiatry UTHealth, <i>Houston, TX</i> • Work with a team of psychiatrists, residents and social workers treating patients with a variety of psychiatric disorders
2017 – 2019	Team Leader, Health Resource Coordinator Charm City Clinic, <i>Baltimore, MD</i> • Assisted clients with a variety of challenges related to medical care, employment, and housing • Trained a new group of volunteers
2016 – 2019	Shadowing, Parkinson's Neuropsychiatric Clinic Johns Hopkins Medicine, <i>Baltimore, MD</i> • Shadowed a psychiatrist treating Parkinson's disease, including psychiatric, cognitive and motor symptoms

- 2015 – 2016 **Emergency Room Volunteer**
Penn Presbyterian Medical Center, *Philadelphia, PA*
- Took incoming calls, paged nurses, restocked supplies, and observed procedures
- 2015 – 2016 **Front-Desk Volunteer**
Washington West Project, *Philadelphia, PA*
- Enrolled patients for STD screening and counseling

Research service

Mentoring

Xiaoke Niu, Jaehwan Kim, Joey Zambelas, Madeline Gomez

Reviewing

Scientific Reports, Frontiers in Computational Neuroscience, Cerebellum

Machine Learning Journal Club

Leadership Committee

Certificates

2023 **Machine Learning Specialization**
Stanford University (Coursera)

Deep Learning Specialization
deeplearning.ai (Coursera)

Organizations

2017 - Now **Society for Neuroscience**

2017 - 2019 **National Ataxia Foundation**

Skills

Electrophysiology: Acute and chronic recording, spike-sorting, eye tracking, receptive field mapping, behavioral task design

MRI: Structural scans

Data analysis: MATLAB, Python, SPSS

Machine Learning: PyTorch, scikit-learn, dimensionality reduction, decoding, regression

Task Development: E-Prime, PsychToolbox, MonkeyLogic

Languages

English: Fluent

Spanish: Basic